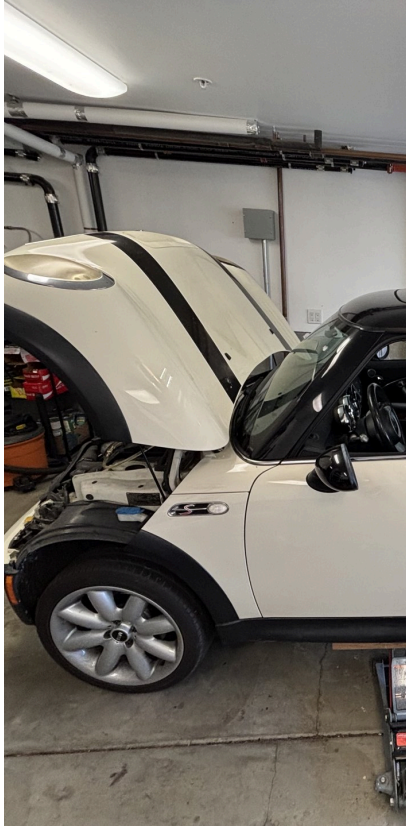
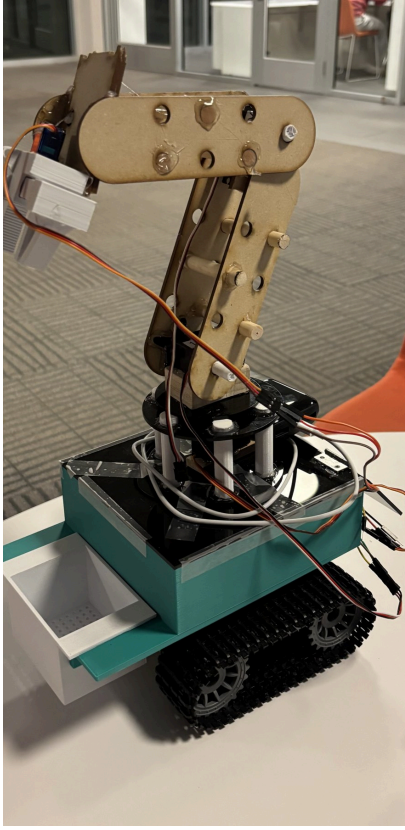






MECHANICAL ENGINEERING PORTFOLIO

Cyrus Chambers

B.S. Mechanical Engineering, Minor in Computer Science · Northeastern University

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The place in the engineering world that fascinates me the most is the intersection of mechanical hardware, software, and physics theory. I love seeing the tangible application of my code and building the mechanical system that it interacts with. This portfolio collects the projects I am most proud of, from vehicle dynamics simulation on a Formula SAE electric race car, to rebuilding the suspension on my own car, to robots I have designed, built, and programmed.

EDUCATION

Northeastern University, Class of 2029
GPA 3.94 · Dean's List
Tau Beta Pi

SEEKING

Mechanical engineering co-op or internship, Spring 2027

INTERESTS

Automotive, space, defense, and robotics hardware

01 · NORTHEASTERN ELECTRIC RACING

Vehicle Dynamics and Steering Hardware

Performance Analysis Engineer · Boston, MA · January 2026 to Present



Northeastern's Formula SAE car. I work on the vehicle dynamics team, on both simulation and parts for the car.

Building a bicycle model

My most recent project for the club was building a bicycle model in Python for the team's simulation software. A bicycle model lumps the two front tires and the two rear tires together, so the car is described by just a front axle, a rear axle, and the mass between them. I built it to follow the car's real specifications. This allowed us to measure the cause and effect of various parameters without the added complexity of nonlinear tire models and load transfer.

Why it was useful

The model let us vary parameters such as weight distribution, wheelbase, and cornering stiffness and see the effect on acceleration, lap time, yaw, and slip angles. For example, whether the car understeers or oversteers depends on only a couple of factors, including front/rear mass distribution and cornering stiffness, which makes it much easier to find which input values create the desired output.

This gave us a better understanding not only of how, but why the car performs the way it does, and it sped up the design process by letting us compare options before committing to physical changes.

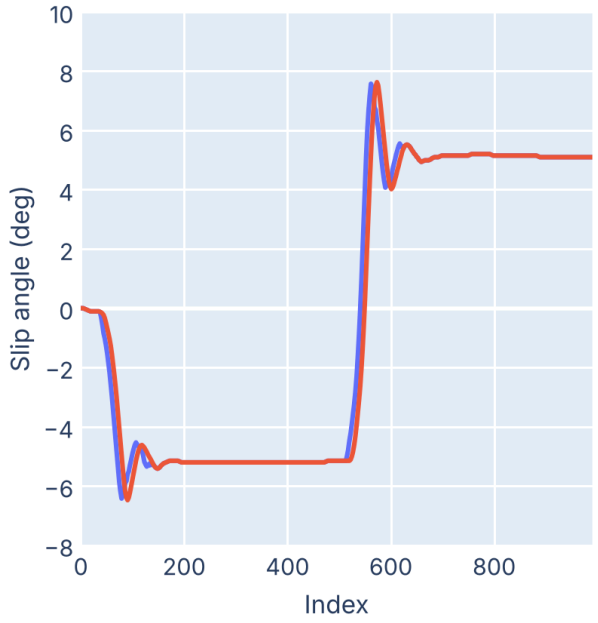
Design and machining. I also do mechanical design and machining on NER. I designed and machined custom parts for the steering column using SolidWorks and manual milling for the fabrication.

Python · Vehicle dynamics · Simulation · SolidWorks · CAD/CAM · Manual milling

Simulation Results

Output plots from the team's simulation software

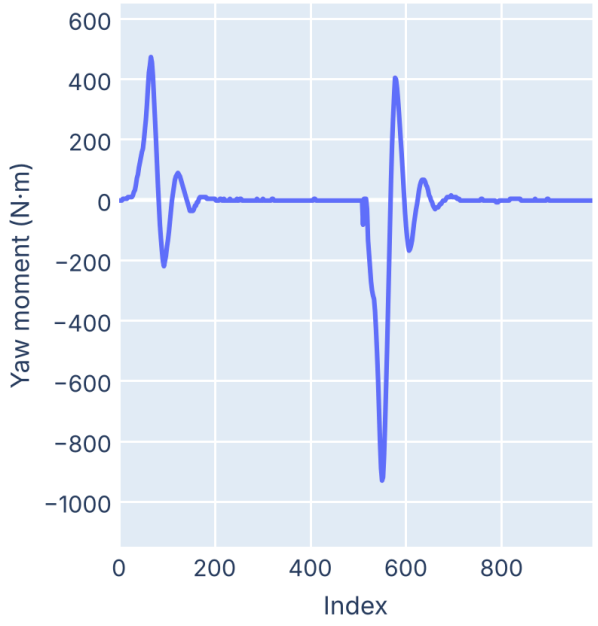
These are plots from my bicycle model. Being able to see slip angles and yaw moment respond as I changed one input at a time is what made the model so effective for comparing design options.



Front axle Rear axle

Front and rear slip angles (deg). The car is given a steering input in one direction and then the other. After a short, well damped transient, both axles settle to a steady slip angle of about 5°.

The front and rear slip angles end up almost equal, which in a bicycle model means the car is close to neutral steer. Moving the mass distribution or the cornering stiffness opens a gap between the two lines, and which axle slips more tells you whether the car will understeer or oversteer.



Yaw moment

Yaw moment (N·m). The yaw moment spikes each time the steering changes, swings back, and then returns to zero as the car settles into the turn.

Returning to zero is what should happen: in a steady corner the front and rear tire forces balance, so there is no net moment rotating the car. How quickly the oscillation dies out shows how stable the car's yaw response is.

02 · AUTOMOTIVE MECHANIC

Rebuilding a 2005 Mini Cooper S

San Francisco, CA · June to September 2026



My R53 Mini Cooper S on jack stands mid-job, with the rear suspension and brakes taken apart.

$5,000+
saved in professional labor charges

~100,000 mi
added to the life of the car

50+ mi
running perfectly since the rebuild

Automotive Mechanical Repair

Another project that I am very proud of is the work I have done on my twenty-year-old manual Mini Cooper S. This past summer I rebuilt the entire suspension and did a full brake job, which included pulling the leaking CV drive axles out of the transmission.

- Rebuilt the full suspension system
- Replaced the ball joints and CV axles
- Full brake job: rotors, pads, sensors, and fluid bleed
- Transmission and engine oil change

Suspension · Brakes · Drivetrain · Hands-on repair

A Challenge I Overcame

During the test drive, after I thought I had successfully replaced both axles, the left axle popped out of the transmission and the car refused to go into gear. Being in San Francisco, I made the quick decision to pull into a turnout with the momentum I had left, so I could bring the car safely to a stop. I then put the car in neutral and rolled it back down the hill to a small side street near my house, where I worked on the axle and fixed it properly.

03 · ENGINEERING: PROJECT-BASED

Beach Cleaning Robot

Northeastern University · Boston, MA · January to April 2026

For my project-based engineering course, our team built a robot on treads with a robotic arm and a detachable bucket, made to pick up trash on the beach and separate the collected debris from the sand.

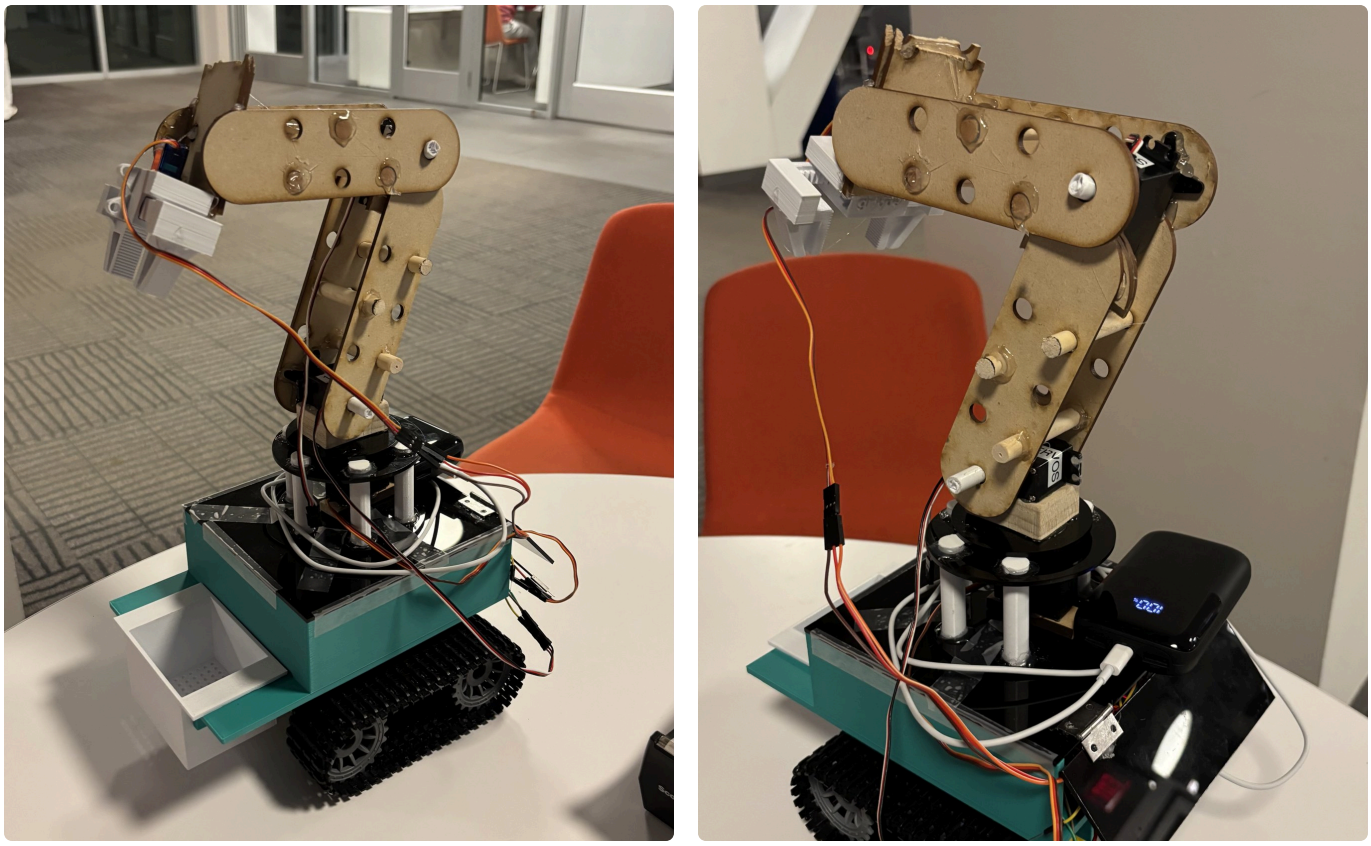
What I did

- Designed and fabricated the robot body and arm components in SolidWorks, using 3D printing, laser cutting, and various fasteners
- Contributed to the circuit design and the programming for servo control of the treads, arm, and claw joints

The arm is made of laser cut links that rotate on a turntable base, with a servo at each joint and a claw at the end. The bucket slides out of the side of the body so the collected trash can be emptied once it is separated from the sand.

I liked this project because it incorporated multiple aspects of engineering: designing parts on the computer, making them with my hands, and then writing the code that makes them move. A lot of things do not fit or work the first time, and working through that with my team was the best part.

SolidWorks · 3D printing · Laser cutting · Servo control · Circuit design

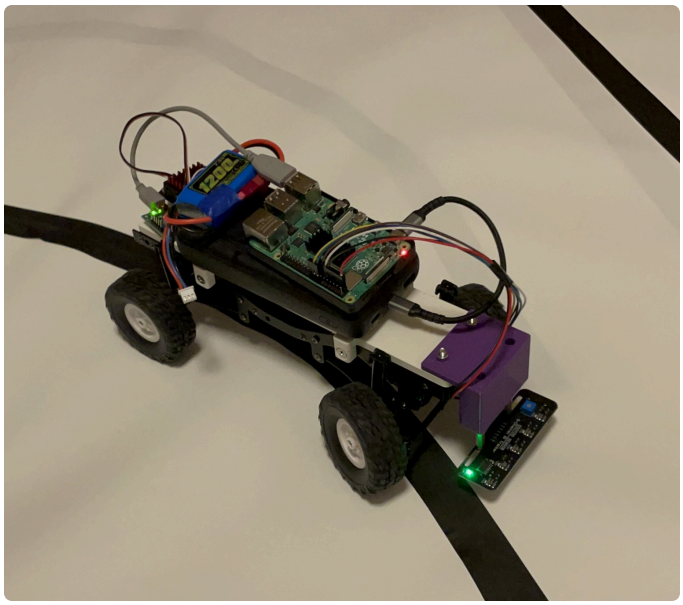


The finished robot: tracked chassis, slide-out bucket, and the laser cut arm on its rotating base.

04 · PERSONAL PROJECT

Autonomous Line Following Car

San Francisco, CA · January to June 2025



The converted RC car on its track, with the Raspberry Pi on top and the IR sensor array at the front.

I converted an RC car into an autonomous vehicle. I wired IR sensors to a Raspberry Pi and wrote the Python code that turned the sensor readings into steering and throttle commands, so the car could follow a black line on its own.

I also included a recovery routine for curves tighter than the car's turning radius. When the car loses the line in a sharp curve, it backs up and takes another angle at it, which let it get around turns it physically could not make in one pass.

Raspberry Pi · Python · IR sensors · Controls

SKILLS & BACKGROUND

How I Work

I have a sustained work ethic and attention to detail and I love to learn new things. I am highly collaborative and an optimist, and these things carry me through the point where something feels like it will never work, to the point where I am able to solve the problem. In all of my projects, I document procedures and results with precision and write clear reports, which have been appreciated by my team members.

SOFTWARE

SolidWorks, AutoCAD, Git, Vim, VSCode, LaTeX, Excel, Adobe Illustrator, Lightroom/Photoshop

MACHINING & TOOLS

Manual milling, 3D printing, laser cutting, soldering, oscilloscope, digital multimeter

LANGUAGES

Python, Java, C++, MATLAB, Bash

COURSEWORK

Calculus I–III, Differential Equations, Linear Algebra, Mechanics, Statics, Material Science, Electricity & Magnetism